

Problem Set #3

Marli Fichtner

September 2025

1 Introduction

This brief analysis uses the 2019 National Survey of Children’s Health (NSCH) to explore differences in age at autism diagnosis and a few related patterns. I focus on 2019 because it is the most recent year in which the NSCH asked “Age at autism diagnosis,” and the goal here is descriptive, exploratory work rather than multi-year trends.

Motivated by recent questions about access to care across places, I compare outcomes for children living in a *principal city* (a large city or urban center in a metro area) versus children living outside principal cities. Unless otherwise noted, results are descriptive and intended to highlight patterns worth exploring further.

I present three visuals:

1. Average household attitudes toward their insurance’s coverage of mental health services, by state.
2. The distribution of age at autism diagnosis for children who live in principal cities versus those who do not.
3. The type of provider who first told the family about the autism diagnosis, by principal-city status.

2 Visualizations

2.1 Visualization 1

Average Attitude towards Mental Health Coverage Across States

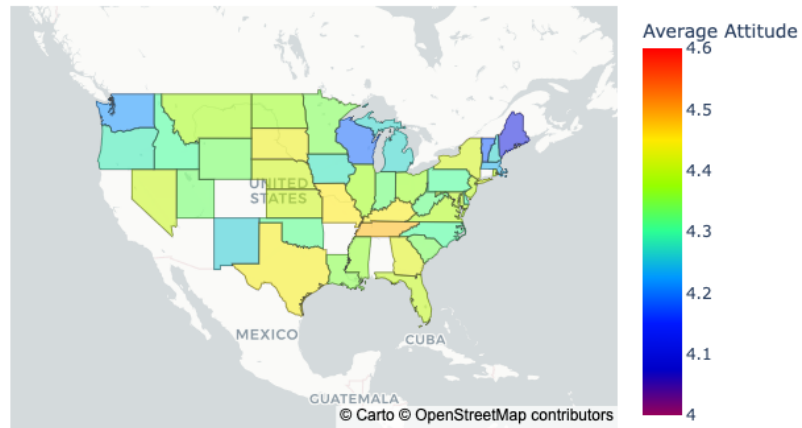


Figure 1: Average household attitude toward insurance coverage of mental health services, by state (NSCH 2019).

Figure 1 shows generally positive views of mental health coverage across states. Scores cluster in a fairly narrow band (roughly 4.0–4.6 on a 1–5 scale), so cross-state differences are modest. Most states fall near 4.3–4.5. Given the compressed range, a future version could pair the map with a simple ranked bar chart to make small differences easier to read.

2.2 Visualization 2

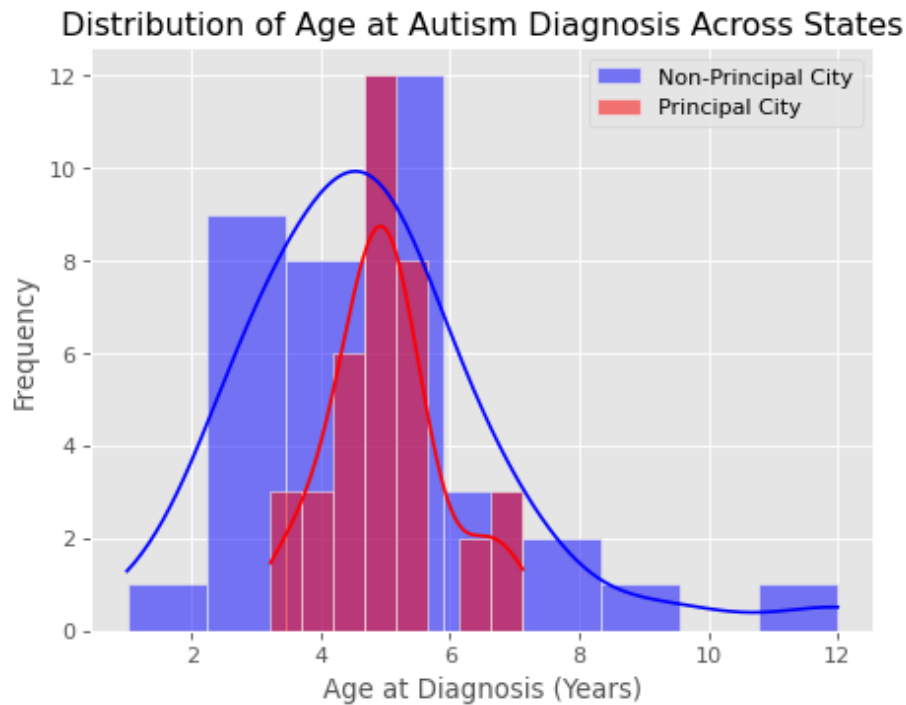


Figure 2: Age at autism diagnosis, principal city vs. non-principal city (NSCH 2019).

Figure 2 overlays the distributions of age at diagnosis for children in principal cities and outside principal cities. Both groups have many diagnoses around ages 4–5. The principal-city distribution appears a bit more concentrated, while the non-principal-city distribution is slightly more spread out and includes some younger ages. These are descriptive patterns but they suggest place-based differences worth a closer look.

2.3 Visualization 3

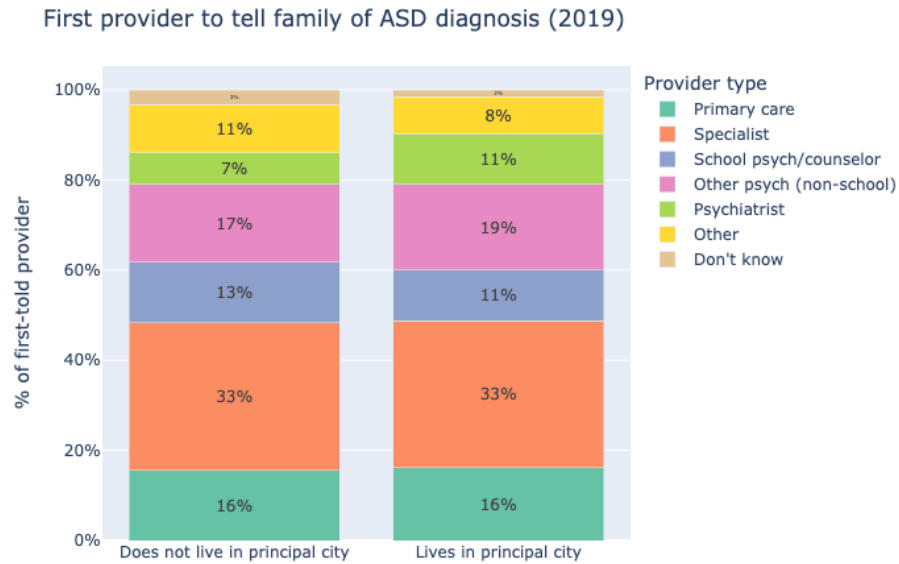


Figure 3: First provider to tell the family about the autism diagnosis, by principal-city status (NSCH 2019).

Figure 3 compares who first communicated the diagnosis. The two groups look broadly similar, with only small shifts across provider types. This similarity suggests that, at least in this sample, the point of first diagnosis communication does not differ much by principal-city status.

3 References

U.S. Census Bureau. (2019). *National Survey of Children's Health (NSCH), 2019 datasets*. United States Census Bureau.